Reinforcement Learning

Mid Semester Exam

23/09/2025

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Instructions: You have about 120 minutes to work on the questions. Answers with no supporting steps will receive no credit. No resources, other than a pen/pencil, are allowed. In case you believe that required information is unavailable, make a suitable assumption.

Question 1. 5 **marks** An agent experiences the following episode, which is provided next as a sequence of (s_t, a_t, r_{t+1}) tuples.

$$(0,1,1),(1,1,2),(2,0,5),(0,0,2),(1,2,-1),(2,1,3),(2,1,2),(1,0,1).$$

Calculate unbiased estimates of the values of states observed during the episode.

Question 2. 20 **marks** A bus has two seats. Each seat can be sold at a price of either INR 1 or INR 2. When no seat has been sold, advertising a price of INR 1, per seat, results in 1 seat being sold with probability 0.3, 2 seats being sold with probability 0.2, and no seats sold otherwise. Also, when no seat has been sold, advertising a price of 2 results in 1 seat being sold with probability 0.2, 2 seats being sold with probability 0.2, and no seats sold otherwise.

When exactly one seat has been sold, advertising a price of 1 results in the sale of the remaining seat with probability 0.6. When exactly one seat has been sold, advertising a price of 2 results in the sale of the remaining seat with probability 0.5. Since the bus has only two seats, prices are advertised only till both seats are sold.

Every time a price is advertised, the bus company incurs a cost of INR 1. The goal is to maximize the expected profit (sum of prices at which the seats are sold minus the total cost of advertising prices) from the sale of the seats. Sketch the MDP. Derive the optimal value function and the optimal policy.

Question 3. 20 **marks** Consider the following policy improvement proposal. Given a policy PMF $\pi(a|s)$, for every action a in the set of actions A and for every state s, we create a policy

$$\pi_+(a_k|s) = \begin{cases} \pi(a_k|s) - \epsilon_k & a_k \neq a_*, \\ \pi(a_k|s) + \sum_{k:a_k \neq a_*} \epsilon_k & a_k = a_*, \end{cases}$$

where a_* is the greedy action in state s (given that the reward-to-go is given by the value function $v_\pi(s)$) and $\epsilon_k \in [0, \pi(a_k|s)]$ for all actions $a_k \neq a_*$. Will the policy improvement policy proposal always result in a policy $\pi+$ that is at least as good as the policy π? Support your claim with a proof.

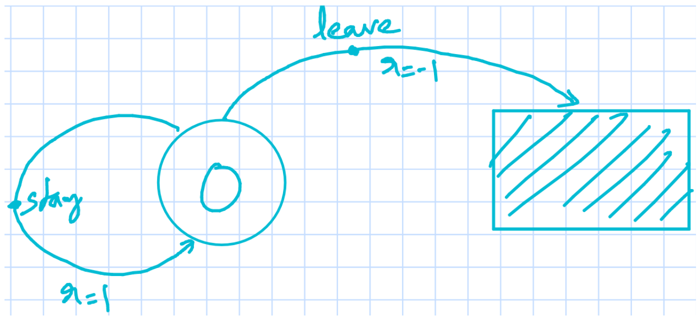

Fig. 1: A simple Markov Decision Process

What about the following proposal? Does it result in an improvement? We set

$$\pi_+(a_k|s) = \begin{cases} 0 & a_k = a_{\text{min}}, \\ \pi(a_k|s) + (1/(K-1))\pi(a_{\text{min}}|s) & \text{otherwise}, \end{cases}$$

where K is the total number of actions and a_{min} is the action at the current time that minimizes the value of state s, given that the reward-to-go is given by the value function $v_\pi(s)$. Essentially, we set the probability of choosing the worst choice of current action to zero and distribute the probability mass assigned to it amongst other actions.

Question 4. 20 **marks** Consider the MDP in Figure 1. An agent interacts with an environment modeled by the MDP and obtains a sequence of rewards $1, 1, 1, -1$ before termination of episode. Calculate an every-visit and a first-visit estimate of the value of state 0. Assume the agent used a policy that picks the action *stay* with probability $p > 0$ and picks *leave* otherwise. Which estimate results in a smaller error, where error is the square of the difference between the estimate and the true value, and when (as a function of p)? Explain why your answer makes sense, given the MDP, the policy, and the provided episode [*Hint*: Think in terms of the lengths of episodes you used to come up with the estimates.].

Question 5. 15 **marks** Consider the MDP in Figure 1. Derive the optimal policy and value function. Impose any required assumptions to ensure a finite optimal value function.

Question 6. 20 **marks** Consider the MDP in Figure 1. An agent interacts with an environment modeled by the MDP and obtains a sequence of rewards $1, 1, 1, -1$ before termination of episode. Assume the agent used a behavior policy $b(a|s)$ that picks the action *stay* with probability $p > 0$ and picks *leave* otherwise to generate the episode.

Consider a target policy μ that picks the action *leave* with probability 1. Estimate the value of the target policy using every-visit Monte Carlo and the given episode.

For the given MDP and behavior and target policies, calculate $E_\mu[G_t|S_t = s]$ given that you can calculate averages only over the behavior policy b. That is rewrite $E_\mu[G_t|S_t = s]$ as $E_b[G'_t|S_t = s]$, where G'_t is an appropriately scaled version of the return random variable G_t. Also calculate $\text{Variance}_b[G'_t|S_t = s]$.

Question 1. 5 **marks** An agent experiences the following episode, which is provided next as a sequence of (s_t, a_t, r_{t+1}) tuples.

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Calculate unbiased estimates of the values of states observed during the episode.

Sample mean together with first-visit MC results in unbiased estimates.

[Choice of estimate 2/5]

The estimates are:

$\tilde{v}(0) = 15$ 1/5

$\tilde{v}(1) = 14$ 1/5

$\tilde{v}(2) = 12$ 1/5

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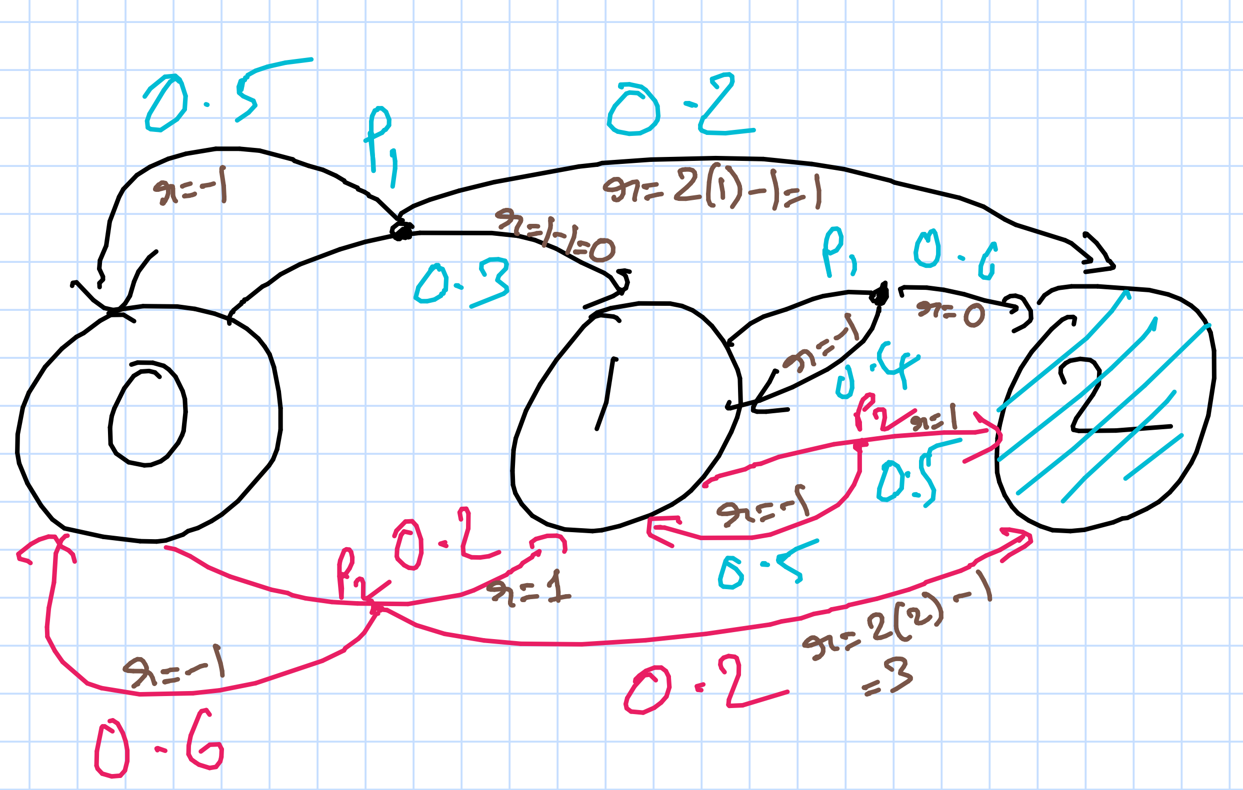


Price $P_1 = 1$
$P_2 = 2$.

Up to 5/20 for the MDP.

The set of states is $\{0, 1, 2\}$. State captures occupancy.

The reward obtained for a transition is the revenue from seat sales (price × number seats sold) − advertising cost (given as −1).

Consider the Bellman Optimality Equations: (Note that $V_*(2) = 0$)

$$V_*(0) = \max\{0.5(-1 + V_*(0)) + 0.2(1 + V_*(2)) + 0.3(0 + V_*(1)),$$
$$0.6(-1 + V_*(0)) + 0.2(3 + V_*(2)) + 0.2(1 + V_*(1))\}$$

$$= \max[-0.3 + 0.5V_*(0) + 0.3V_*(1),$$
$$0.2 + 0.6V_*(0) + 0.2V_*(1)\}$$

Up to 5/20

$$V_*(1) = \max\{0.4(-1 + V_*(1)) + 0.6(0 + V_*(2)),$$
$$0.5(-1 + V_*(1)) + 0.5(1 + V_*(2))\}$$

$$= \max\{-0.4 + 0.4V_*(1), 0.5V_*(1)\}$$

$$= 0.5V_*(1)$$

Up to 5/20

$\Rightarrow V_*(1) = 0$. Also, optimal action is INR 2.

$$\therefore V_*(0) = \max\{-0.3 + 0.5V_*(0), 0.2 + 0.6V_*(0)\}$$

$$= 0.2 + 0.6V_*(0)$$

$$\Rightarrow 0.4V_*(0) = 0.2$$

$V_*(0) = 0.5$. Also, optimal action is INR 2.

Calculation of optimal values Up to 2.5/20

Optimal policy 2.5/20

To summarize:

The optimal value function is

$$V_*(0) = 0.5,\ V_*(1) = 0,\ V_*(2) = 0.$$

The optimal policy π_* is:

$$\pi_*(0) = 2,\ \pi_*(1) = 2.$$

Question 3. 20 **marks** Consider the following policy improvement proposal. Given a policy PMF $\pi(a|s)$, for every action a in the set of actions A and for every state s, we create a policy

$$\pi_+(a_k|s) = \begin{cases} \pi(a_k|s) - \epsilon_k & a_k \neq a_*, \\ \pi(a_k|s) + \sum_{k:a_k\neq a_*} \epsilon_k & a_k = a_*, \end{cases}$$

where a_* is the greedy action in state s (given that the reward-to-go is given by the value function $v_\pi(s)$) and $\epsilon_k \in [0, \pi(a_k|s)]$ for all actions $a_k \neq a_*$. Will the policy improvement policy proposal always result in a policy $\pi+$ that is at least as good as the policy π? Support your claim with a proof.

What about the following proposal? Does it result in an improvement? We set

$$\pi_+(a_k|s) = \begin{cases} 0 & a_k = a_{\min}, \\ \pi(a_k|s) + (1/(K-1))\pi(a_{\min}|s) & \text{otherwise}, \end{cases}$$

where K is the total number of actions and $a_{\min}$ is the action at the current time that minimizes the value of state s, given that the reward-to-go is given by the value function $v_\pi(s)$. Essentially, we set the probability of choosing the worst choice of current action to zero and distribute the probability mass assigned to it amongst other actions.

Given $\pi_+(a_k|s) = \begin{cases} \pi(a_k|s) - \epsilon_k & a_k \neq a_*, \\ \pi(a_k|s) + \sum_{k:a_k\neq a_*} \epsilon_k & a_k = a_*, \end{cases}$

We want to confirm whether or not $v_{\pi_+}(s) \geq v_\pi(s)$ for every s.

Note that

$$v_\pi(s) = E_\pi\left[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid S_t = s\right]$$

$$= \sum_a \sum_r \sum_{s'} (r + \gamma v_\pi(s'))\, p(s', r|s,a)\, \pi(a|s)$$

$$= \left[\sum_r \sum_{s'} (r + \gamma v_\pi(s'))\, p(s', r|s, a_*)\right] \pi(a_*|s) + \sum_{a\neq a_*} \left\{\sum_r \sum_{s'} (r + \gamma v_\pi(s'))\, p(s', r|s,a)\right\} \pi(a|s)$$

$$= q_\pi(s, a_*)\pi(a_*|s) + \sum_{a\neq a_*} q_\pi(s,a)\pi(a|s)$$

Showing this inequality Up to 10/20

Note that $q_\pi(s, a_*) \geq q_\pi(s,a)$ for every $a \neq a_*$.

$$\therefore\ v_\pi(s) = q_\pi(s, a_*)\pi(a_*|s) + \sum_{a\neq a_*} q_\pi(s,a)\pi(a|s)$$

$$\leq q_\pi(s, a_*)\left(\pi(a_*|s) + \sum_{k:a_k\neq a_*} \epsilon_k\right) + \sum_{a\neq a_*} q_\pi(s,a)(\pi(a|s) - \epsilon_a)$$

$$= T_{\pi_+} v_\pi(s).$$

We have $T_{\pi_+} v_\pi(s) \geq v_\pi(s)$ for every state s.

Repeated application of T_{π_+} on the left hand side, preserves the ordering w.r.t the right hand side $v_\pi(s)$.

Further, repeated application of T_{π_+} to the L.H.S gives $v_{\pi_+}(s)$.

$$\therefore\ v_{\pi_+}(s) \geq v_\pi(s), \text{ for every } s.$$

π_+ guarantees policy improvement.

The policy evaluation argument Upto 5/20

The second part follows in a similar manner. Note that

$$v_\pi(s) = q_\pi(s, a_*)\pi(a_*|s) + \sum_{a\neq a_*} q_\pi(s,a)\pi(a|s)$$

$$= q_\pi(s, a_{min})\pi(a_{min}|s) + \sum_{a\neq a_{min}} q_\pi(s,a)\pi(a|s)$$

$$\leq \sum_{a\neq a_{min}} q_\pi(s,a)\left(\pi(a|s) + \frac{1}{K-1}\pi(a_{min}|s)\right).$$

This is because $q_\pi(s, a_{min}) \leq q_\pi(s,a)$, for every $a \neq a_{min}$.

We are effectively assigning higher probabilities to all actions better than a_{min} & a prob of 0 to a_{min}.

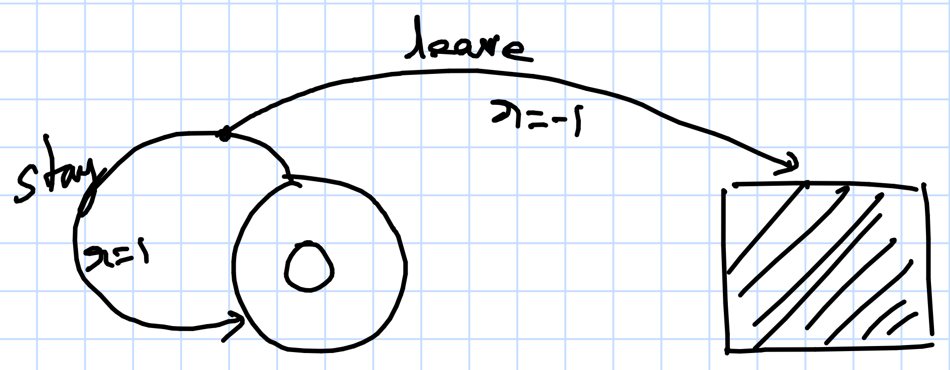
We have $v_\pi(s) \leq T_{\pi_{++}} v_\pi(s)$ for every s.

Rest of the argument is the same as that for part 1.

This proposed policy also guarantees improvement.

Up to 5/20

Question 4. 20 **marks** Consider the MDP in Figure 1. An agent interacts with an environment modeled by the MDP and obtains a sequence of rewards $1, 1, 1, -1$ before termination of episode. Calculate an every-visit and a first-visit estimate of the value of state 0. Assume the agent used a policy that picks the action *stay* with probability $p > 0$ and picks *leave* otherwise. Which estimate results in a smaller error, where error is the square of the difference between the estimate and the true value, and when (as a function of p)? Explain why your answer makes sense, given the MDP, the policy, and the provided episode [*Hint*: Think in terms of the lengths of episodes you used to come up with the estimates.].



Every-visit:

Given the episode $1,1,1,-1$, we can get the returns

$1+1+1-1,\ 1+1-1,\ 1-1,\ -1$.

That is $2, 1, 0, -1$.

The value of the state can be estimated as the sample mean $\frac{2+1+0-1}{4} = \frac{2}{4} = \frac{1}{2}$

Up to $\frac{3}{20}$

First-visit:

We can only use the return 2. Therefore the estimate of the value of the state is 2.

Up to $3/20$

Consider the policy (call it π) that picks stay with probability $p > 0$.

$$v_\pi(0) = p\left(1 + v_\pi(0)\right) + (1-p)(-1)$$

$$= p - 1 + p + p\, v_\pi(0)$$

$$v_\pi(0) = \frac{2p-1}{1-p}.$$

This works for $p \neq 0$. For $p=1$, we will need $\gamma < 1$.

Up to $\frac{5}{20}$

Error in every-visit $= \left(\frac{1}{2} - \frac{2p-1}{1-p}\right)^2$

in first-visit $= \left(2 - \frac{2p-1}{1-p}\right)^2$

Up to $\frac{4}{20}$

$p = 0 \Rightarrow v_\pi(0) = -1$

$p = \frac{1}{4} \Rightarrow v_\pi(0) = \frac{-1/2}{3/4} = -\frac{1}{2}\cdot\frac{4}{3} = -\frac{2}{3}$

$p = \frac{1}{2} \Rightarrow v_\pi(0) = 0$

$p = \frac{2}{3} \Rightarrow v_\pi(0) = \frac{2(2/3)-1}{1-2/3} = \frac{1/3}{1/3} = 1.$

$p = \frac{3}{4} \Rightarrow v_\pi(0) = \frac{2(3/4)-1}{1-3/4} = \frac{1/2}{1/4} = 2.$

As p increases, we expect the value to increase.
These are just some examples.
The smallest value is -1. As p increases, we go past $\frac{1}{2}$ and then 2, which are the every-visit & first-visit estimates, respectively.

Note that v_π increases from -1 to 2 & so on, as p increases from 0 to 1.
Also, the p for which first-visit becomes better than every visit, is a p at which the true value that is greater than $\frac{1}{2}$, which is the every-visit estimate. At our p of interest:

$$\left(\frac{2p-1}{1-p} - \frac{1}{2}\right) = \left(2 - \frac{2p-1}{1-p}\right)$$

$$\Rightarrow 2\left(\frac{2p-1}{1-p}\right) = 2 + \frac{1}{2} = \frac{5}{2}$$

$$2p - 1 = \frac{5}{4}(1-p)$$

$$2p - 1 = \frac{5}{4} - \frac{5p}{4}$$

$$p\left(2 + \frac{5}{4}\right) = \frac{9}{4}$$

$$p\left(\frac{13}{4}\right) = \frac{9}{4}$$

$$\Rightarrow p = \frac{9}{13}$$

For $p \in \left(0, \frac{9}{13}\right)$, every-value is better.

$p \in \left(\frac{9}{13}, 1\right)$, first-value is better.

$p = \frac{9}{13}$, they are equally good.

Figuring the behavior of estimates as a function of p: Up to $\frac{4}{20}$

As p increases, the probability of episodes with fewer stays before leave reduces. Longer episodes (more 1s before -1) are more likely. The first-visit return is from a longish episode. The every-visit returns include returns from episodes as short as of length 1.

$\frac{1}{20}$

Question 5. 15 **marks** Consider the MDP in Figure 1. Derive the optimal policy and value function. Impose any required assumptions to ensure a finite optimal value function.

Note that staying helps accumulate +ve rewards. Leaving results in a reward of -1.

Intuitively, staying is optimal. However, we end up with an infinite sequence of $+1$ (s). The resulting return is ∞, for $\gamma = 1$.

To get around this, we must assume a discount factor $0 \leq \gamma < 1$.

Observing that we require $\gamma < 1$.
Up to 5/15

$$v_*(0) = \max \{ 1 + \gamma v_*(0), -1 \}$$

We must pick between

$$v_*(0) = -1 \text{ and}$$

$$v_*(0) = 1 + \gamma v_*(0)$$

$$\Updownarrow$$

$$v_*(0) = \frac{1}{1-\gamma} > 0 \text{ for all } 0 \leq \gamma < 1.$$

$$\therefore \; v_*(0) = \frac{1}{1-\gamma}.$$

Calculating the optimal
Up to 10/15

Question 6. 20 **marks** Consider the MDP in Figure 1. An agent interacts with an environment modeled by the MDP and obtains a sequence of rewards $1, 1, 1, -1$ before termination of episode. Assume the agent used a behavior policy $b(a|s)$ that picks the action *stay* with probability $p > 0$ and picks *leave* otherwise to generate the episode.

Consider a target policy μ that picks the action *leave* with probability 1. Estimate the value of the target policy using every-visit Monte Carlo and the given episode.

For the given MDP and behavior and target policies, calculate $E_\mu[G_t|S_t = s]$ given that you can calculate averages only over the behavior policy b. That is rewrite $E_\mu[G_t|S_t = s]$ as $E_b[G'_t|S_t = s]$, where G'_t is an appropriately scaled version of the return random variable G_t. Also calculate Variance$_b[G'_t|S_t = s]$.

Every-visit MC gives us the returns (from the behavior policy)

2 corresponding to 1, 1, 1, −1
1 corresponding to 1, 1, −1
0 corresponding to 1, −1
−1 corresponding to −1.

The target policy picks leave with probability 1.
Therefore, all returns other than −1 are reduced to 0.
As regards −1, we must scale it by $\frac{\pi(\text{leave}|0)}{b(\text{leave}|0)} = \frac{1}{1-p}$.

$\therefore$ The corresponding return sample for the target is $\frac{-1}{1-p}$.
We have returns $0, 0, 0, \frac{-1}{1-p}$.
The estimate of the value of the target is $\frac{-1}{4(1-p)}$ [Why does this make sense??]

?

Up to 10/20

Consider $E_b[G'_t|S_t=0]$:

Note that episodes that have one or more stay actions before leave result in $G'_t = 0$.
The episode that has just the reward −1 results in $G'_t = \frac{-1}{1-p}$. This occurs with probability that the behavior picks leave as the first action, which is $(1-p)$.

$$\therefore \; E_b[G'_t|S_t=0] = \left(\frac{-1}{1-p}\right)(1-p) = -1.$$

?

Up to 7/20

$$E_b[G'^2_t|S_t=0] = \left(\frac{-1}{1-p}\right)^2 (1-p) = \frac{1}{1-p}.$$

$$\therefore \; \text{Var}[G'_t|S_t=0] = \frac{1}{1-p} - (-1)^2 = \frac{1}{1-p} - 1 = \frac{1-1+p}{1-p}$$
$$= p/1-p.$$

Up to 3/20